Brake Pedal Design Documentation

Aditya Pravar Dhavala – 03/21/2026

Objective:

This document details the design of a brake pedal in compliance with the regulations defined in the 2026 Baja SAE rules section B.7.1.1. The brake pedal should be an all-aluminum or steel construction, withstand 2000 N of force without failing, and remain operational in the event of one master cylinder circuit failing. A bias bar system was selected for simplicity and ease of adjusting braking feel to improve dynamics. The pedal is constructed out of 6061 T6 Aluminum Alloy, with a lever ratio of 4:1. The pedal is 8" from pivot to footpad and 2" from pivot to the pushrod attachment point, which gives us our 4:1 ratio (8/2 = 4).

Setups

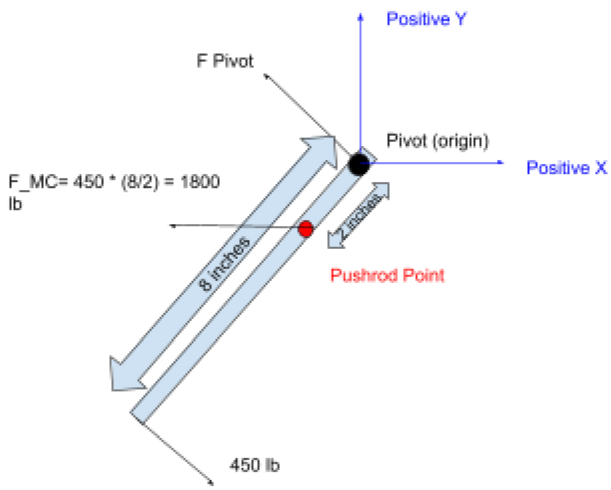

Figure 1: Free Body Diagram

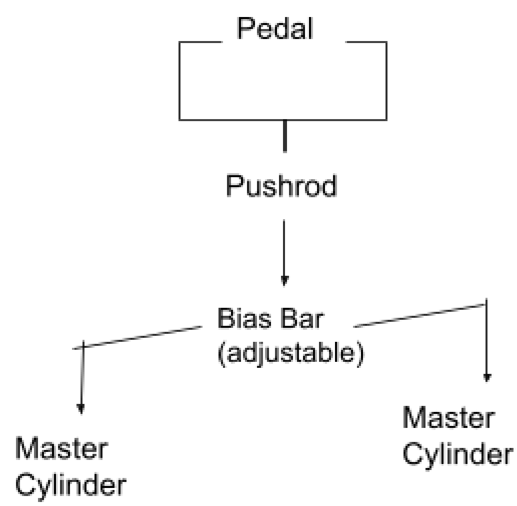

Figure 2: Bias Bar Brake Setup

Calculations

Section Modulus

1.5" wide, 0.75" thick at pivot and pushrod

$$S = (w * h^2) / 6 = (1.5 * 0.75^2) / 6 = .14625 \text{ in}^3$$

Force

The moment caused by slamming on this 8 inch brake pedal with 450 lb of force is 3600 lb in. 450 lb represents a worst case panic scenario as defined in the Baja SAE rule book Section B.7.1.1

$$M = L\,x\,F = 450 \text{ lb} * 8 \text{ in} = 3600 \text{ lb*in}$$

Bending Stress

6061 T6 Aluminum was chosen for its significant weight savings over 4130 Chromoly Steel (0.0975 lb/in^3 vs 0.284 lb/in^3), despite having a lower yield strength (42,000 psi vs 63,000 psi).

$\sigma = M/S$ = 3600 lb*in / 0.14625 in^3 = 24,615 psi
Yield strength of 6061 T6 Aluminum: 42,000 psi

Safety Factor

Assuming a baseline safety factor of 1.5 for a high-load panic scenario, the designed brake will be safe for Baja SAE when faced with a sudden 450 lb force.

SF = Yield Strength / σ
SF = 42000 / 24615 = 1.706
1.706 > 1.5000 → Safe

Finite Element Analysis

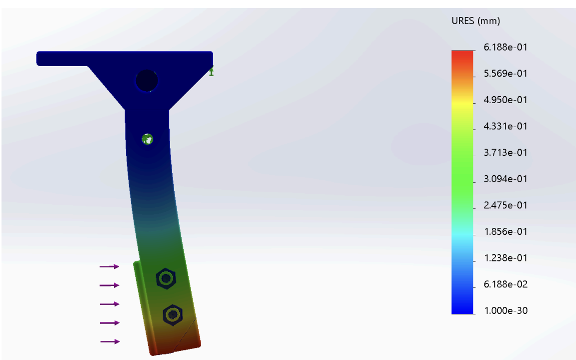

Figure 3: Displacement

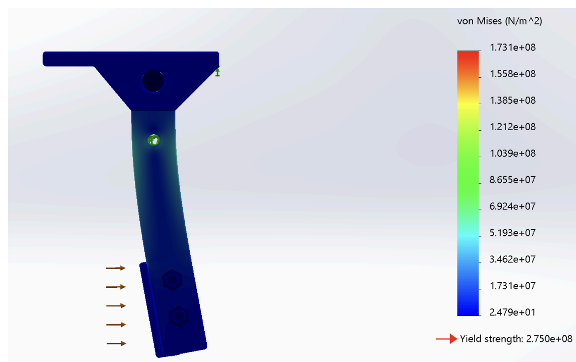

Figure 4: Stress

The FEA plots show very light strain and a maximum displacement of around 6/10ths of a millimeter, indicating that the pedal can handle the emergency scenario 450 lb force. The Stress diagram shows a max stress of 1.731 x 10^8 N/m^2, which translates to roughly 25,106 psi. The calculated and simulated maximum stress values strongly agree with each other, supporting accuracy of calculations.

Unobstructed Travel

Calculations performed assuming usage of 2x Wilwood Master Cylinder No. 260-10372 and Calipers sourced from Baja SAE 2018 example parts list. Assumed setup consists of 2x front calipers and 1 rear caliper. A running clearance (rc) of 0.020” was also used as a conservative estimate to ensure proper runtime.

Front Calipers: Single Piston, 2” Bore
Rear Caliper: Dual Piston: 1” Bore
Master Cylinder Bore: ¾”
Master Cylinder Stroke: 1.12”

Pushrod Pin X at Start: -0.96
Pushrod Pin X at End: +1.8”
Total Pushrod X Travel: 1.8 - (-0.96) = 2.76”

The pushrod connects via clevis and clevis pin, and has enough travel in the X direction to comfortably traverse the entire stroke of the master cylinder, ensuring that the system will engage. It also has significant overhead, more than double the stroke of one cylinder, so in the event one circuit fails, it will be able to continue forward until it engages the remaining working circuit regardless of bias bar adjustment.

Area of Master Cylinder: $A = \frac{\pi d^2}{4}$ = .750^2 *π / 4 = 0.4418 in^2
Volume of Master Cylinder: V = A * Stroke = .442 * 1.12 = 0.492 in^3

Front volume: $V_f = \frac{\pi}{4} b^2 * (rc) * (calipers * pistons/caliper)$ = π / 4 * 2.0^2 * 0.020 * 2 = 0.1257 in^3
Front stroke: Fs = 0.1257/0.4418 = 0.285 in
Front Margin: Fm = 1.12 - 0.285 = 0.835 in > 0

Rear volume: V_r = π / 4 * 1.0^2 * 0.020 * 2 = 0.0314 in^3
Rear stroke: Rs = 0.0314/0.4418 = 0.071 in
Rear Margin: Rm = 1.12 - 0.071 = 1.049 in > 0

Both circuits have significant positive margins, indicating that the circuit engages before the full stroke is traversed. Since the brake design can hinge further than the full stroke, and the full stroke far exceeds the necessary stroke, it is safe to assume that the brake can operate unobstructed.

Initial and final positions of pushrod pin hole:

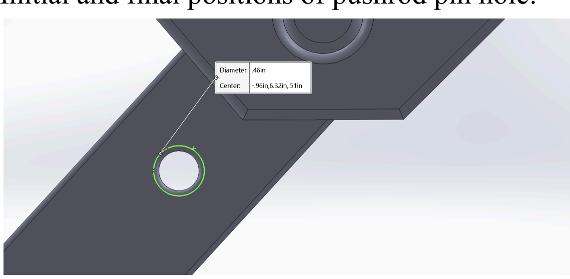

Figure 5:(X:-0.96in, Y: 6.32in, Z: 0.51in)

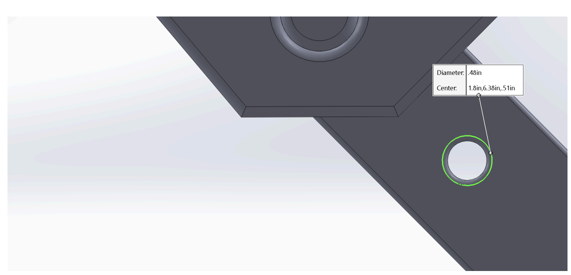

Figure 6:(X: 1.8in, Y: 6.38in, Z: 0.51in)

Sketches

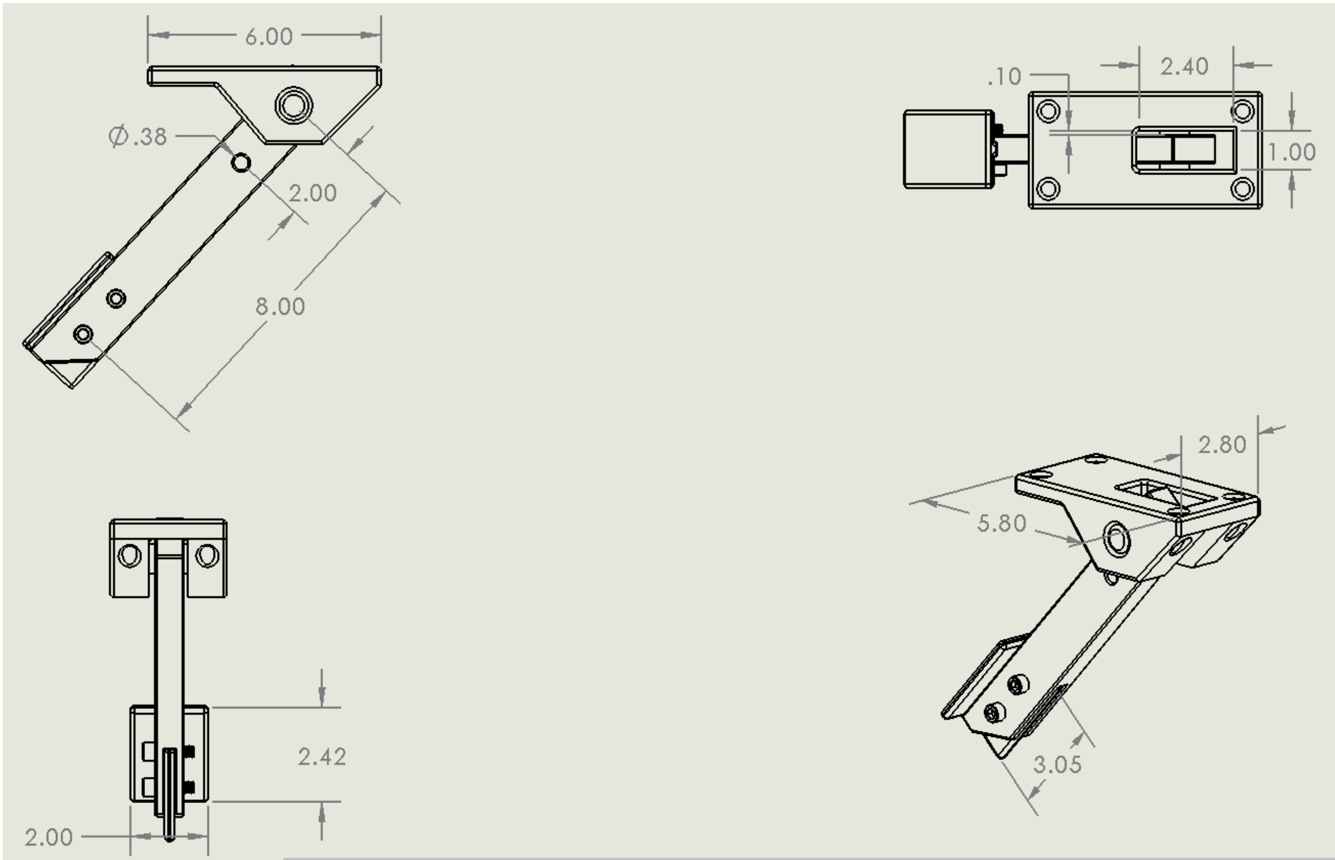

Figure 7: Solidworks Sketches